SEM – 3

III - SEMESTER (3 Core Papers/2 Elective Papers)

Code	Title of the Paper	Category of Courses	Teaching Hours Per Week			No. of Credits	Sessional Marks CIE	Semester End marks SEE	Exam Duration	Max Marks
			L	T	P					
MSP 301	Artificial Intelligence and Deep Learning	DSC	3	1	0	4	30	70	3 Hrs	100
MSP 302	Cloud Computing - AWS / Azure / GCP	DSC	3	1	0	4	30	70	3 Hrs	100
MSP 303	Soft Skills & Personality Development	SEC	3	1	0	4	30	70	3 Hrs	100
MSP 304	Elective – III	DSE	3	1	0	4	30	70	3 Hrs	100
MSP 305	Elective – IV	DSE	3	1	0	4	30	70	3 Hrs	100
OEC 306	Open Elective	OEC	3	1	0	4	30	70	3 Hrs	100
MSP 307	Hands-on: Elective – III	-	0	0	4	2	30	70	3 Hrs	100
MSP 308	Hands-on: Elective - IV	-	0	0	4	2	30	70	3 Hrs	100
Total						**28**				**800**

SEMESTER III

45

MSP 301 - Artificial Intelligence and Deep Learning

(Semester-III of MS Programme)

Department	Computer Science	Course Type	MSP
Course Title	Artificial Intelligence and Deep Learning	**Course Code**	MSP 301
L-T-P	3-1-0	**Credits**	4
Contact Hours	60 Hrs	**Duration of SEE**	3 Hrs
SEE Marks	70	**CIE Marks**	30

Unit I:

Introduction to Machine Learning - Intro to ML, Types of Machine Learning algorithms, Supervised, Unsupervised, Reinforcement, ML Applications, ML Life Cycle, challenges in Machine Learning, ML vs AI, ML vs Deep Learning.

Unit II:

Intro to AI - Basics of Artificial Intelligence, Applications of AI, Types of AI, AI agents, Artificial Neural Networks, Fuzzy Logic, Introduction to Genetic Algorithm, Soft Computing.

Unit III:

Intro to Deep Learning - Fundamentals of Deep Learning, Architecture of Deep Learning, Types of Deep Learning Networks – FFNN, RNN, CNN, Restricted Boltzmann Machine, Autoencoders, Deep Learning Applications, Limitations, Advantages, Disadvantages.

Unit IV:

Introduction to AI Networks with Keras - Biological neurons, multilayer perceptron's, and back-propagation, multilayer perceptron using Keras and visualize the runs and graphs using Tensorboard, Training Deep Neural Networks - Reusing pre-trained layers, using faster optimizers and avoiding overfitting by regularization.

Unit V:

Processing Sequences using RNNs & CNNs - Recurrent Neural Networks - RNN predict the future, the problem they face like limited short-term memory and solutions to these problems - LSTM (Long Short-Term Memory) and GRU cells, NLP Concepts – NLP Techniques. 45

References:

1. S. Russel and P. Norvig, "Artificial Intelligence – A Modern Approach", Pearson Education. 2. B. Yegnanarayana, Artificial Neural Networks, Prentice Hall of India. 3. Neural Networks and Fuzzy Logic System by Bart Kosko, PHI Publications

MSP 302 - Cloud Computing - AWS / Azure / GCP
46
(Semester-III of MS Programme)

Department	Computer Science	Course Type	DSCC
Course Title	Cloud Computing - AWS / Azure / GCP	Course Code	MSP 302
L-T-P	3-1-0	Credits	4
Contact Hours	60 Hrs	Duration of SEE	3 Hrs
SEE Marks	70	CIE Marks	30

Unit I:
Basics of Cloud Computing – Intro to Cloud Computing, Architecture, Cloud Technologies, Types of Cloud, Cloud Service Models, Cloud Service Providers, Cloud Server, Cloud Computing Applications, Security Risks of Cloud Computing.

Unit II:
Introduction to AWS – AWS Infrastructure, AWS Free tier, AWS IAM – User, Roles, Policies, EC2 – AWS EC2 & Pricing, EBS, EBS Volume, Security Group, AWS AMI, AWS Load Balancing - Create Load Balancer, Types of Load Balancers – Application Load Balancer, Network Load Balancer, Classic Load Balancer, Auto Scaling, Cloud Watch.

Unit III:
AWS Storage Service – S3 Buckets, Objects, Keys, Regions, Create S3 Bucket, AWS Versioning, Lifecycle Management, S3 Transfer, Application Services – AWS SQS, AWS SNS, API Gateway, AWS VPC – Custom VPC, NAT Gateways, AWS Route S3, AWS SFTP – Create Server, Users, File Transfer with SFTP, AWS EKS, AWS Lambda Function.

Unit IV:
46 Basics of GCP - Intro to GCP, Building Blocks – Compute, Storage, Network, Identity Management, Services – Databases, Data & Analytics, AI & ML, Developers Tools, Enterprise Services.

Unit V:
Basics of Azure – Intro to Azure, Azure Portal, Storage Services – Azure Storage account, Blob Storage, Storage Security, File Storage, Storage Monitoring, Compute Services – Azure VM Storage, VM Security, Monitoring, Network Services - Virtual Network, Network Security, Availability Zones & Sets, Azure Load Balancer, App Services – Azure Web App, Azure Mobile App, Azure API.

References:

1. “Cloud Computing for Dummies” by Judith S. Hurwitz, Fern Halper, Robin Bloor, Marcia Kaufman. 2. “AWS: The Complete Beginner's Guide” by Stephen Baron.

MSP 303-Soft Skills & Personality Development

(Semester-III of MS Programme) 47

Department	Computer Science	Course Type	DSCC
Course Title	Soft Skill & Personality Development	**Course Code**	MSP 303
L-T-P	3-1-0	**Credits**	4
Contact Hours	60 Hrs	**Duration of SEE**	3 Hrs
SEE Marks	70	**CIE Marks**	30

Unit-I

Introduction to Soft-Skills-Personal Skills: Knowing Oneself/Self-Discovery-Confidence Building- Defining Strengths- Developing Positive Attitude- Thinking Creatively- Improving Perceptions -Forming Values.

Unit-II

Interpersonal and Social Skills: Understanding Others-Developing Inter-personal relationship- Team Building-Group Dynamics-Networking-Problem-solving.

Unit-III

Communication Skills: Art of Listening-Art of Speaking-Art of Reading-Art of Writing-Art of Writing E-mails: Email etiquette 47

Unit-IV

Corporate Skills: Working with others- Developing a proper body language-behavioral etiquettes and mannerism- Time Management –Stress Management

Unit-V

Job-hunting skills: Writing Resume/CV- Interview Skills-Group Discussion-Mock interviewMock GD-Goal Setting-Career Planning

References:

1. Meena K and V. Ayothi (2013) A Book on Development of Soft Skills (Soft Skills: A Road Map to Success), P. R. Publishers & Distributors, No. B-20 & 21, V. M. M Complex, Chatiram Bus Stand, Tiruchirappalli-620002. (Phone No: 0431-2702824Mobile No.: 9443370597, 9843074472) 2. Alex K. (2012) Soft Skills-Know Yourself & Know the World, S. Chand & Company LTD, Ram Nagar, New Delhi-110055. Mobile No.: 9442514814 (Dr.K.Alex)

MSP 304 B- Data Science with PySpark

(Elective-III of MS Programme)

Department	Computer Science	Course Type	DSCC
Course Title	Data Science with PySpark	**Course Code**	MSP 304 B
L-T-P	3-1-0	**Credits**	4
Contact Hours	60 Hrs	**Duration of SEE**	3 Hrs
SEE Marks	70	**CIE Marks**	30

Unit I:
Intro to Python & Pyspark - Basics of Python, Programming with RDD: Foreach Loop, Using Reduce Function, Viewing Records from MySQL, Spark Joins, Pyspark Joins Examples, Word Count

Unit II:
Data Frames Essentials - Intro to Data Frames, Read, Write, validate &explore Clean, Manipulate, Join, Aggregate.

Unit III:
Classification of Spark MLlib - Intro to Spark MLlib, Model Selection and Tuning in MLlib, Data Formatting and Transformations, Train and Evaluate Models, Train & Test Models, Train & Test Models [Multilayer PC], Train & Test Models [Naive Bayes], Train & Test Models [Linear SVM], Train & Test Models [Decision Tree], Train & Test Models [Random Forest] 57

Unit IV:
Natural Language Processing - Intro to NLP, Feature Transformers, Feature Extractors, Data Prep, Vectorize, Train & Eval, Regression in MLlib, Regression in Pyspark MLlib, Data Prep, Linear Regression, Decision Tree Regression, Random Forest Regression, Gradient Boosted Tree Regression

Unit V:
MLlib - Introduction to Clustering, K-Means & Bisecting K-Means in MLlib code, Gaussian Mixture Modeling in MLlib, Frequent Pattern Mining in MLlib, Intro to Spark Structured Streaming, Intro to Streaming Data Using Sockets.

References:

1. "Learning Pyspark" by Tomasz Drabas Denny 2. "Learning Spark: Lightning-Fast Data Analytics" by Jules S. Damji

305 A-AI and Deep Learning through TensorFlow

(Elective-IV of MS Programme)

Department	Computer Science	Course Type	DSCC
Course Title	AI and Deep Learning through TensorFlow	**Course Code**	MSP 305 A
L-T-P	3-1-0	**Credits**	4
Contact Hours	60 Hrs	**Duration of SEE**	3 Hrs

Unit I:
Intro to AI - Basics of Artificial Intelligence, Applications of AI, Types of AI, AI agents, Artificial Neural Networks, Fuzzy Logic, Introduction to Genetic Algorithm, Soft Computing.

Unit II:
Intro to Deep Learning - Fundamentals of Deep Learning, Architecture of Deep Learning, Types of Deep Learning Networks – FFNN, RNN, CNN, Restricted Boltzmann Machine, Autoencoders, Deep Learning Applications, Limitations, Advantages, Disadvantages.

Unit III:
TensorFlow - What is TensorFlow, Installation Through pip, Installation Through conda, Architecture of TensorFlow, Advantage & Disadvantage of TensorFlow, Basics of TensorFlow – Representation of a Tensor, Types of Tensors, TensorFlow Perception – Single Layer, Hidden Layer, Multilayer Perceptions.

Unit IV:
ANN in TensorFlow – Intro, Implementation of NN, Classification of NN, Liner Regression, CNN in TensorFlow – Intro, Working of CNN, Training of CNN, MNIST Dataset in CNN, RNN in TensorFlow – Intro, Working of RNN, RNN Time Series, LSTM RNN in TensorFlow.

Unit V:
Case Studies Case 1: CNN for Digit Recognition Case 2: CNN for Breast Cancer Detection Case 3: CNN for Predicting the Bank Customer Satisfaction Case 4: CNN for Credit Card Fraud Detection Case 5: RNN - LSTM for IMDB Review Classification Case 6: Google Stock Price Prediction with RNN and LSTM

References:

1. "Hands-on Machine learning with Scikit-Learn, Keras, and TensorFlow" by Aurelion Geron. 2. "Deep Learning with TensorFlow 2 and Keras: Regression, ConvNets, GANs, RNNs, NLP, and More with TensorFlow 2 and the Keras API" by Amita Kapoor, Antonio Gulli, and Sujit Pal.

306 - Digital Marketing

Unit-I

Principles and Drivers of New Marketing Environment - Web 2.0- Digital Media Industry -Reaching Audience Through Digital Channels- Traditional and Digital Marketing Introduction to Online Marketing Environment - Dotcom Evolution - Internet Relationships -Business in Modern Economy Integrating E-Business to an Existing Business Model Online Marketing Mix SoLoMo (Social-Local Mobile)- Social Media Sites & Monetisation- Careers in Social Media Marketing, Online content development & key word optimisation.

Unit-II

Purchase Behaviour of Consumers in Digital Marketing Format Online Customer Expectations - Online B2C Buying Process Online B2B Buying Behaviour -Social Media Marketing Segments- Forms of Search Engines Working of Search Engines - Revenue Models in Search Engine Positioning - SEO - Display Advertising - Trends, Web Analytics.

Unit-III

Product Attributes and Web Marketing Implications Augmented Product Concept Customizing the Offering - Dimensions of Branding Online Internet Pricing Influences -Price and Customer Value Online Pricing Strategies and Tactics Time-based Online Pricing Personalized Pricing Bundle Pricing- Internet Enabled Retailing - Turning Experience Goods into Search Goods

Unit-IV

Personalization through Mass Customization - Choice Assistance - Personalized Messaging -Selling through Online Intermediaries Direct to Customer Interaction Online Channel Design for B2C and B2B Marketing- Integrating Online Communication into IMC Process -Online Advertising - Email Marketing - Viral Marketing - Affiliate Marketing -Participatory Communication Networks Social Media Communities Consumer Engagement - Co-Created Content Management-Interactive Digital Networks Customer Led Marketing Campaigns.

Unit-V

Role of Social Media- Social Community (Facebook, Linkedin, Twitter etc.)- Social Publishing (Blog, Tumblr, Instagram,Pinterest, Wikipedia, Stumble Upon etc.)- Social Entertainment (YouTube, MySpace, Flickr etc.)- Social Commerce (Trip Advisor, 4 Squares, Banjo etc.) Social Media Measurement& Metrics- Data Mining and social media- Role of social media in Marketing Research- Big Data and social media- Crowd Sourcing- Legal and Ethical aspects related to Digital Marketing. Marketing in Metaverse.